Electricity Price Spike Forecasting with Deep Learning

Martin He
Carnegie Mellon University
Pittsburgh, PA
kunzeh@andrew.cmu.edu

Abstract

This project studies deep learning methods for forecasting electricity prices in the ERCOT market, with a particular emphasis on rare but economically significant price spikes. Classical baselines, including ARIMA-family models, will be compared with sequence-based neural architectures, specifically LSTM and, if time permits, N-BEATSx, incorporating exogenous weather variables from NOAA. Model performance will be evaluated using MAE and RMSE across the full distribution, as well as spike-conditioned metrics and quantile loss to assess tail-risk prediction. The goal is to determine whether modern sequence-based deep learning models can improve robustness and extreme-event sensitivity relative to classical time-series baselines in volatile electricity markets.

1 Motivation and Objectives

Electricity prices exhibit strong seasonality, weather dependence, and heavy-tailed spikes that make forecasting challenging. Accurate prediction of extreme price movements is important not only for forecasting accuracy in the average case, but also for grid reliability, risk management, and energy trading decisions.

This project aims to:

1. Establish rigorous ARIMA-family baseline models.
2. Develop deep learning forecasters, particularly LSTM and N-BEATSx, that incorporate exogenous weather features.
3. Evaluate overall forecasting performance and spike-specific behavior using standard and tail-aware metrics.

The central research question is:

> *Do modern sequence-based deep learning models provide measurable improvements in electricity price spike prediction over classical time-series baselines?*

2 Related Work

Electricity price forecasting is challenging due to non-stationarity, multiple seasonal patterns, weather dependence, and occasional extreme price spikes. Classical statistical approaches such as ARIMA models remain strong baselines for short-horizon forecasting, particularly when seasonal and autoregressive structures dominate the dynamics.

Neural sequence models such as LSTM have been widely applied to time-series forecasting because they can capture nonlinear temporal dependencies. More recent architectures such as N-BEATSx

Preprint. Under review.

extend neural forecasting frameworks by incorporating exogenous covariates, which can improve performance in electricity price prediction where weather variables influence demand and price volatility.

3 Methodology

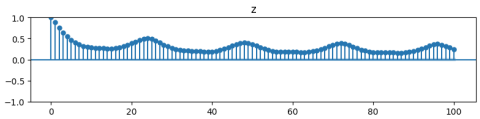
3.1 Baseline Model

$$\phi(B)(1-B)^d y_t = \theta(B)\varepsilon_t$$

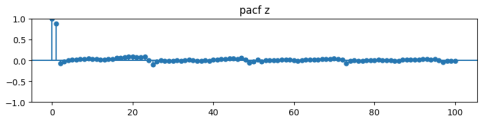
An ARIMA model was implemented as the primary statistical baseline. Before fitting the model, the electricity price series was transformed using the inverse hyperbolic sine (asinh) transformation to reduce the impact of extreme price spikes while preserving sign and relative scale.

$$y_t = \operatorname{asinh}(x_t) = \ln\left(x_t + \sqrt{x_t^2 + 1}\right),$$

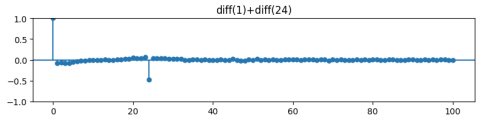
Autocorrelation (ACF) and partial autocorrelation (PACF) plots were then examined to determine appropriate differencing. First-order differencing was applied to remove the overall trend, and an additional seasonal difference with lag 24 was used to account for daily seasonality in the hourly data.



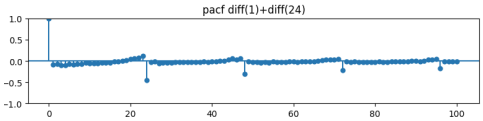
(a) ACF of transformed series



(b) PACF of transformed series



(c) ACF after differencing



(d) PACF after differencing

Figure 1: ACF and PACF diagnostics before and after differencing

The ACF of the raw series shows slow decay and strong seasonal peaks at multiples of 24 hours, indicating non-stationarity and daily periodicity. After applying first-order and seasonal differencing, the ACF decays rapidly with a single significant spike at lag 1 and a seasonal spike at lag 24, suggesting an MA(1) structure with seasonal MA(1) components. The PACF does not show a clear cutoff, indicating weak autoregressive effects.

3.2 Deep Learning Models (yet to be implemented)

The primary deep learning models are:

- **LSTM**: a recurrent sequence model with configurable depth, hidden size, dropout, and input horizon
- **N-BEATSx**: a neural basis expansion model with exogenous covariates and interpretable trend / seasonality decomposition

The deep learning portion of the project will investigate whether exogenous weather variables improve both average-case prediction and sensitivity to rare spikes.

4 Dataset

4.1 Dataset

The study uses publicly available electricity price data from the ERCOT Houston Hub. The dataset consists of hourly locational marginal prices (LMPs), which capture the cost of supplying electricity at a specific location in the ERCOT wholesale market.

The dataset spans four years from January 2020 through December 2023. To preserve the temporal structure of the data, the series was divided chronologically into training, validation, and test sets:

- Training set: January 2020 – December 2021
- Validation set: January 2022 – December 2022
- Test set: January 2023 – December 2023

Before modeling, the price series was transformed using the inverse hyperbolic sine transformation to mitigate the influence of extreme price spikes. The model is currently trained on the differenced series, specifically the first difference and a seasonal difference with lag 24, to remove trend and daily seasonality. As a result, the modeled variable represents changes in electricity prices rather than the absolute price level.

The resulting dataset forms a stationary hourly time series suitable for ARIMA-based forecasting and subsequent comparison with deep learning models.

5 Evaluation and Loss Functions

5.1 Loss Functions

Model performance is evaluated using Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE). For a set of predictions $\hat{y}_t$ and observations y_t, these metrics are defined as

$$\text{MAE} = \frac{1}{n}\sum_{t=1}^{n} |y_t - \hat{y}_t|$$

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{t=1}^{n} (y_t - \hat{y}_t)^2}$$

Let y_t denote the differenced price series. A spike event is defined relative to the empirical 95th percentile of the absolute historical changes. Let

$$q_{0.95} = \text{Quantile}_{0.95}(|y_1|, |y_2|, \ldots, |y_T|)$$

denote the 95th percentile of the absolute values of the historical series up to time T. A spike event occurs when

$$|y_t| > q_{0.95}.$$

5.2 Evaluation

Model performance is evaluated using standard forecasting metrics including Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE). MAE measures the average magnitude of prediction errors, while RMSE penalizes larger deviations more heavily.

To better understand model behavior during extreme price movements, evaluation is conducted separately for two regimes: a normal regime and a spike regime. The spike regime is defined using a threshold on the absolute value of the transformed price series. Observations with $|y_t|$ greater than the threshold are classified as spikes, while the remaining observations belong to the normal regime.

Dataset	Regime	MAE	RMSE
Validation	Normal	0.2129	0.3259
Validation	Spike	2.1738	2.4160
Test	Normal	0.2461	0.3569
Test	Spike	2.2575	2.5963

Table 1: ARIMA(0,0,1) forecasting performance on differenced price series.

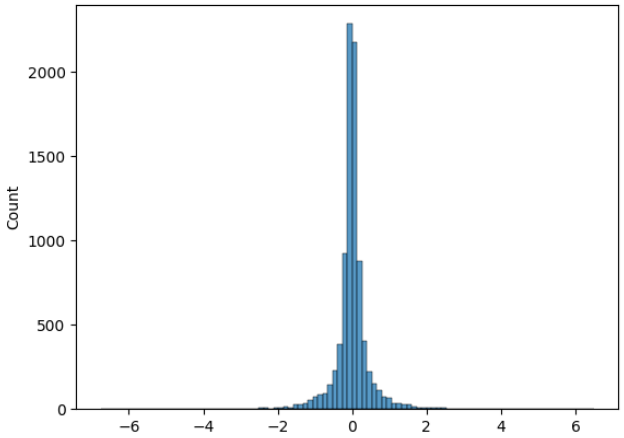


Figure 2: Distribution of the transformed and differenced electricity price series

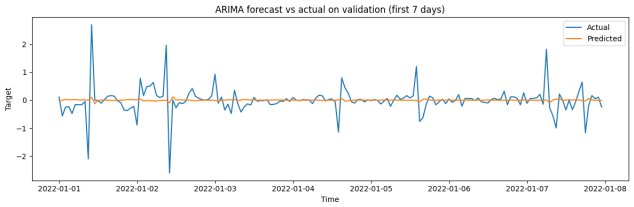


Figure 3: ARIMA forecast compared with actual values on the validation set (7-day window)

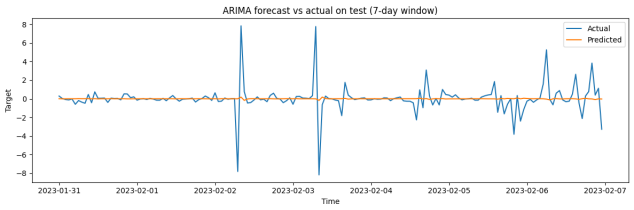


Figure 4: ARIMA forecast compared with actual values on the test set (7-day window)

The results indicate that the ARIMA baseline performs well in normal price regimes but exhibits significantly larger errors during spike periods, highlighting the difficulty of modeling extreme price movements with linear time-series models.

The slightly worse performance on the test set may reflect shifts in market dynamics, potentially driven by weather variability, demand changes, or other external factors, which can introduce distributional changes that reduce forecasting accuracy

6 Experimental Depth

The experiments focus on evaluating a classical ARIMA baseline for electricity price forecasting and analyzing its performance under both normal and spike regimes.

The ARIMA model was trained on the transformed and differenced price series using a walk-forward forecasting procedure. At each step, the model produces a one-step-ahead prediction based on the available historical data. This setup mimics real-world forecasting conditions where future observations are unavailable at prediction time.

To examine the robustness of the model, evaluation was conducted on both validation and test datasets spanning multiple years of hourly electricity prices. Performance was assessed using MAE and RMSE, computed separately for normal price movements and spike events defined using the 95th percentile threshold.

Future experiments will extend this baseline comparison by training deep learning models such as LSTM and N-BEATSx on the same dataset. These models will be evaluated using the same forecasting setup and metrics to determine whether nonlinear architectures improve spike prediction performance relative to classical time-series models.

7 Risk and Feasibility

One challenge in electricity price forecasting is the presence of rare but extreme price spikes. These events are difficult to model using linear time-series approaches such as ARIMA and can lead to large prediction errors. As shown in the baseline experiments, the ARIMA model performs well under normal price movements but struggles during spike regimes.

Another potential challenge is the limited predictability of electricity price dynamics. External factors such as weather conditions, demand fluctuations, and market disruptions can introduce structural changes in the time series that are not captured by simple statistical models.

Despite these challenges, the project remains feasible within the course timeline. A working ARIMA baseline has already been implemented, including preprocessing, walk-forward forecasting, and evaluation under both normal and spike regimes. Future work will focus on training deep learning models such as LSTM and N-BEATSx using the same dataset and evaluation framework, allowing direct comparison with the baseline model.

8 Scope and Contribution

This project focuses on evaluating forecasting methods for electricity prices in the ERCOT Houston Hub market, with particular attention to rare but economically significant price spikes. The current work establishes a statistical baseline using an ARIMA model trained on transformed and differenced price data, together with a forecasting pipeline that includes walk-forward prediction and evaluation under both normal and spike regimes.

The primary contribution of this work is an empirical comparison framework for assessing how different forecasting approaches perform under volatile electricity market conditions. In particular, the project evaluates forecasting accuracy not only under typical price movements but also during extreme price changes defined using a 95th-percentile spike threshold.

Future stages of the project will extend this framework by training deep learning models such as LSTM and N-BEATSx on the same dataset. These models will be evaluated using the same experimental setup and metrics, enabling a direct comparison between classical statistical models and modern neural forecasting approaches.

9 Next Steps

Implement the rest of the models as stated

References

[1] R. Weron, *Electricity Price Forecasting: A Review of the State-of-the-Art with a Look into the Future*, International Journal of Forecasting, vol. 30, no. 4, pp. 1030–1081, 2014.

[2] Y. Zhu, R. Dai, G. Liu, Z. Wang, and S. Lu, *Power Market Price Forecasting via Deep Learning*, IEEE Conference Paper, 2018.

[3] K. G. Olivares, C. Challu, G. Marcjasz, R. Weron, and A. Dubrawski, *Neural Basis Expansion Analysis with Exogenous Variables: Forecasting Electricity Prices with NBEATSx*, International Journal of Forecasting, 2022.

[4] V. Sridharan, M. Tuo, and X. Li, *Wholesale Electricity Price Forecasting Using Integrated Long-Term Recurrent Convolutional Network Model*, Energies, vol. 15, no. 7606, 2022.